

Welcome

CS 472/572
Unsupervised Machine Learning

Clustering 2

Week-3 -Class -2

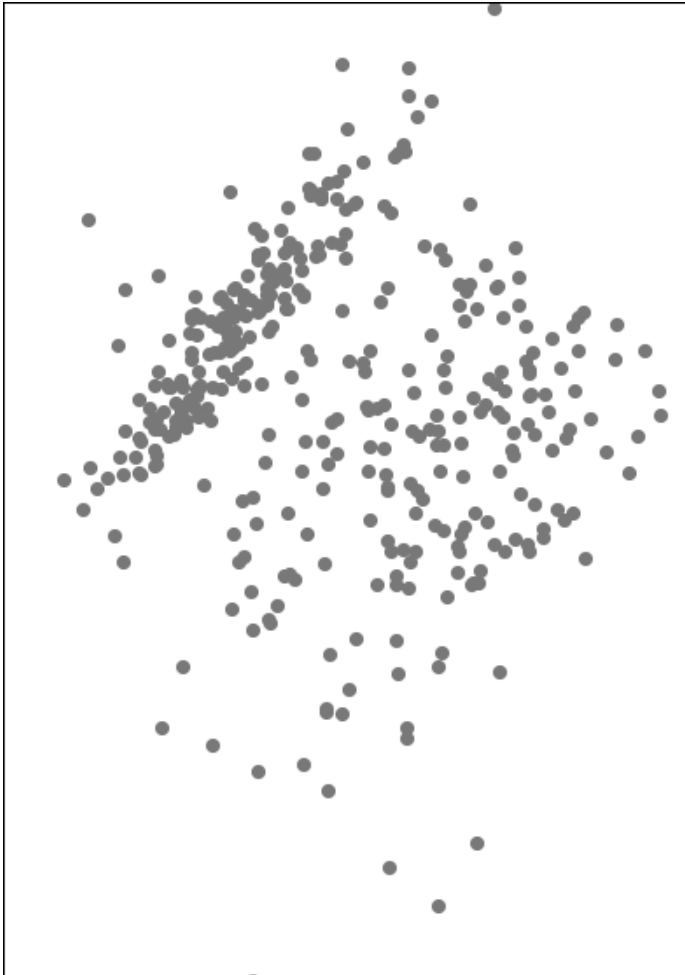
Fall 2025

Instructor: Dr. Khondker Fariha Hossain

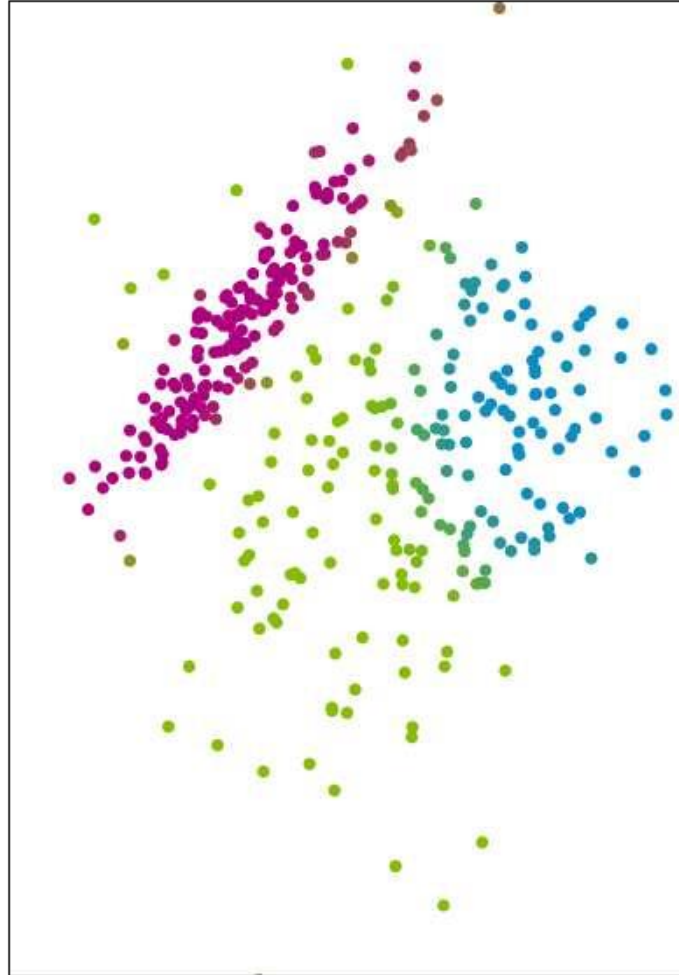
Inferring soft assignments with expectation maximization (EM)

Inferring cluster labels

Data

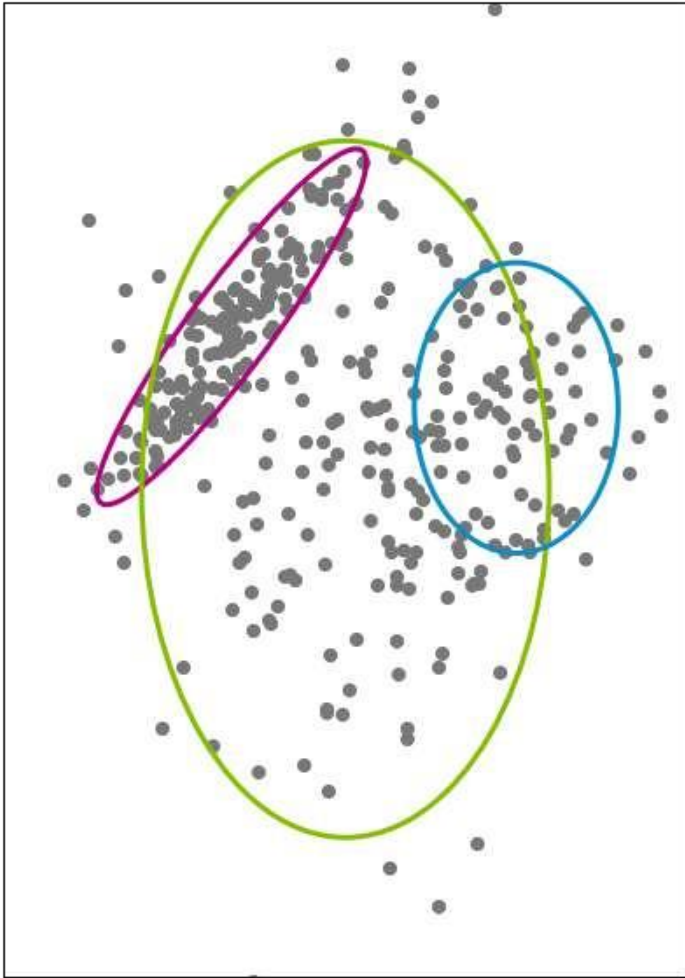


Desired soft assignments



Part 1:
What if we knew the cluster
parameters $\{\pi_k, \mu_k, \Sigma_k\}$?

Compute responsibilities



....

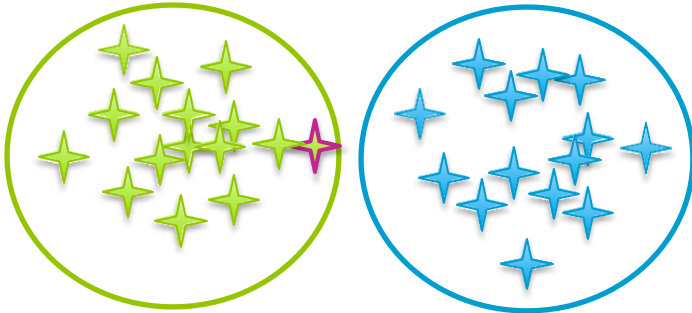
Responsibility cluster k takes for observation i

$$r_{ik} = p(z_i = k \mid \{\hat{\mu}_j, \mu_j, \hat{\sigma}_j\}_{j=1}^K, x_i)$$

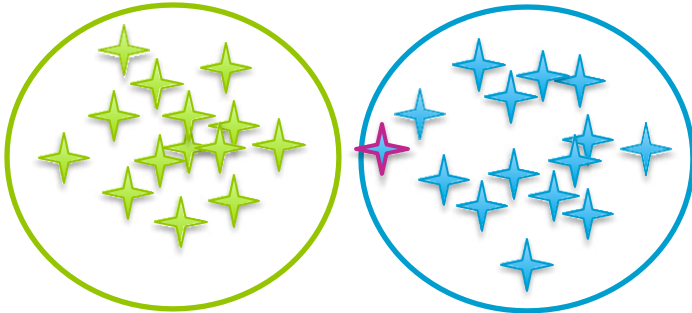
probability of assignment to cluster k

given model parameters and observed value

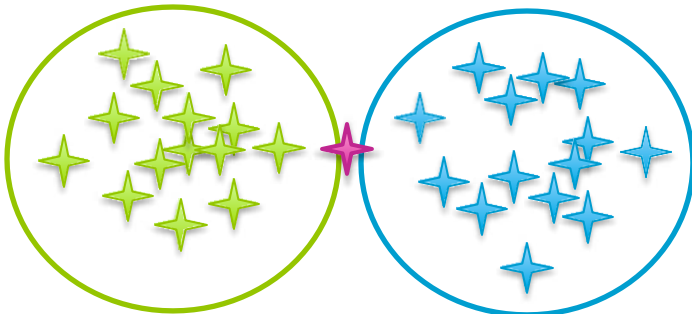
Responsibilities in pictures



Green cluster
takes more
responsibility



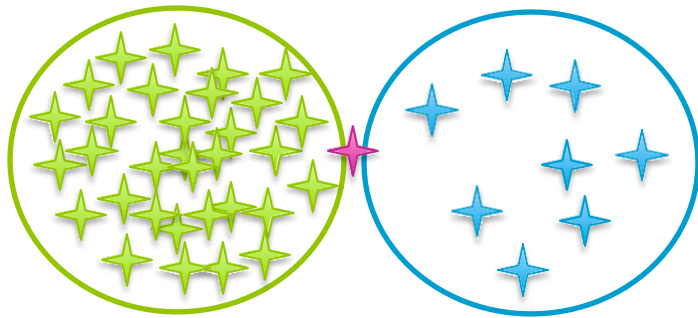
Blue cluster
takes more
responsibility



Uncertain...
split
responsibility

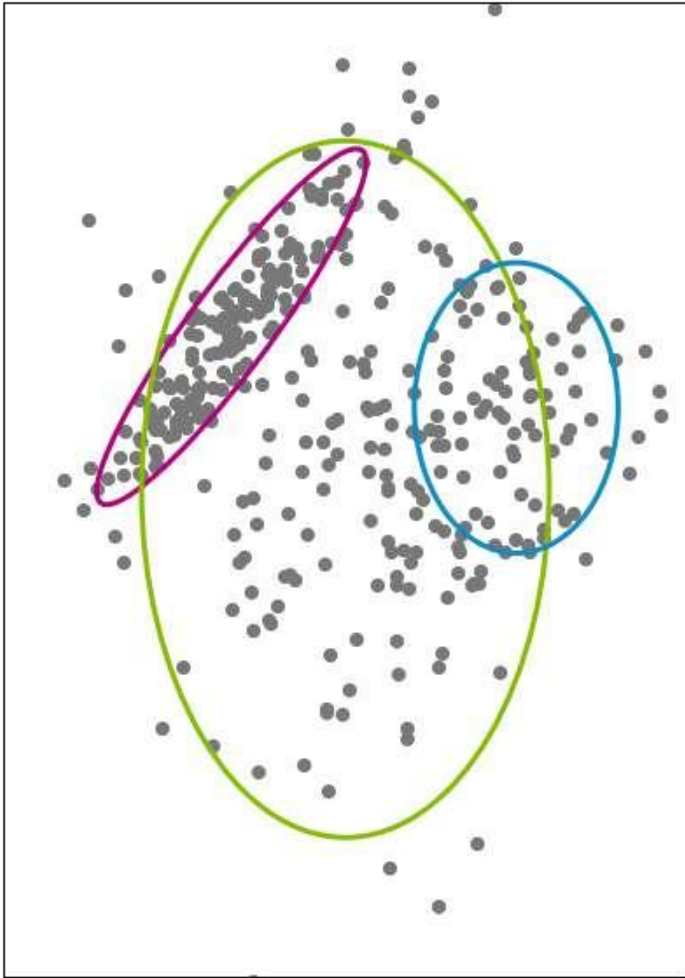
Responsibilities in pictures

Need to weight by cluster probabilities,
not just cluster shapes



Still **uncertain**,
but **green** cluster seems
more probable...
takes more responsibility

Responsibilities in equations

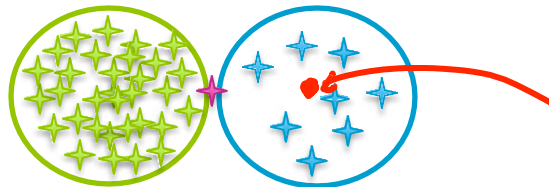


Responsibility cluster k takes for observation i

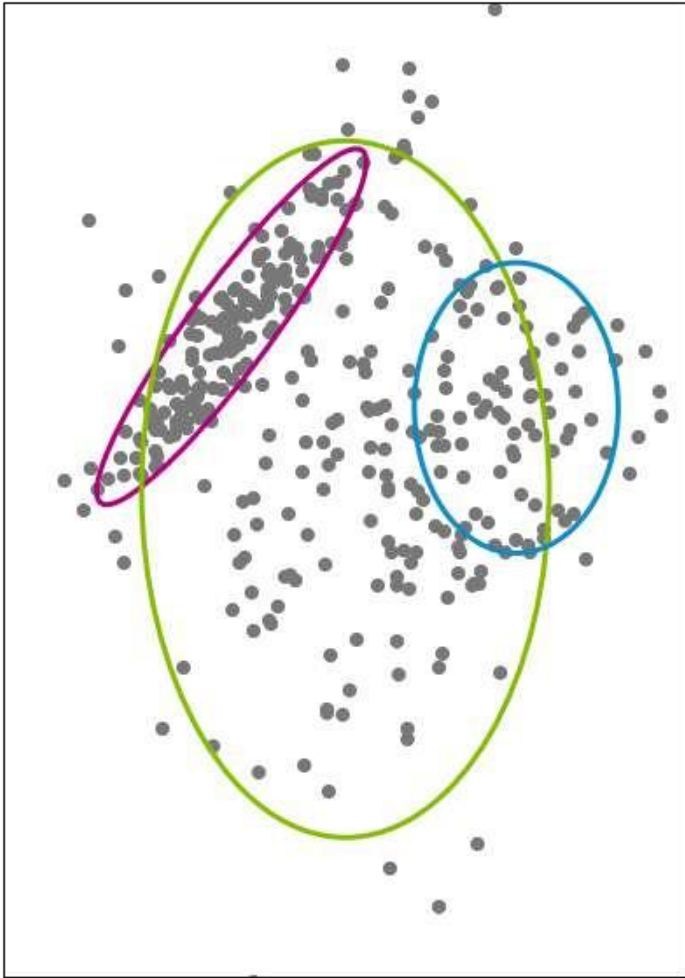
$$r_{ik} = \frac{1}{n_k} N(x_i | \mu_k, \Sigma_k)$$

Initial probability of being from cluster k

How likely is the observed value x_i under this cluster assignment?



Responsibilities in equations



Responsibility cluster k takes for observation i

$$r_{ik} = \frac{\hat{\pi}_k N(x_i | \mu_k, \hat{\Sigma}_k)}{\sum_{j=1}^K \hat{\pi}_j N(x_i | \mu_j, \hat{\Sigma}_j)}$$

Normalized over all possible cluster assignments

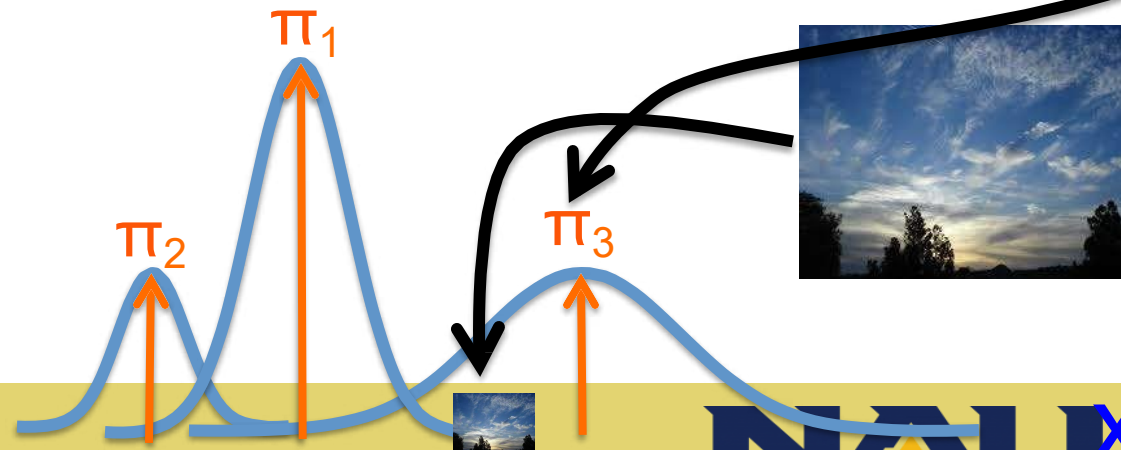
Recall: According to the model...

Without observing the image content, what's the probability it's from cluster k ? (e.g., prob. of seeing "clouds" image)

$$p(z_i = k) = \hat{\pi}_k$$

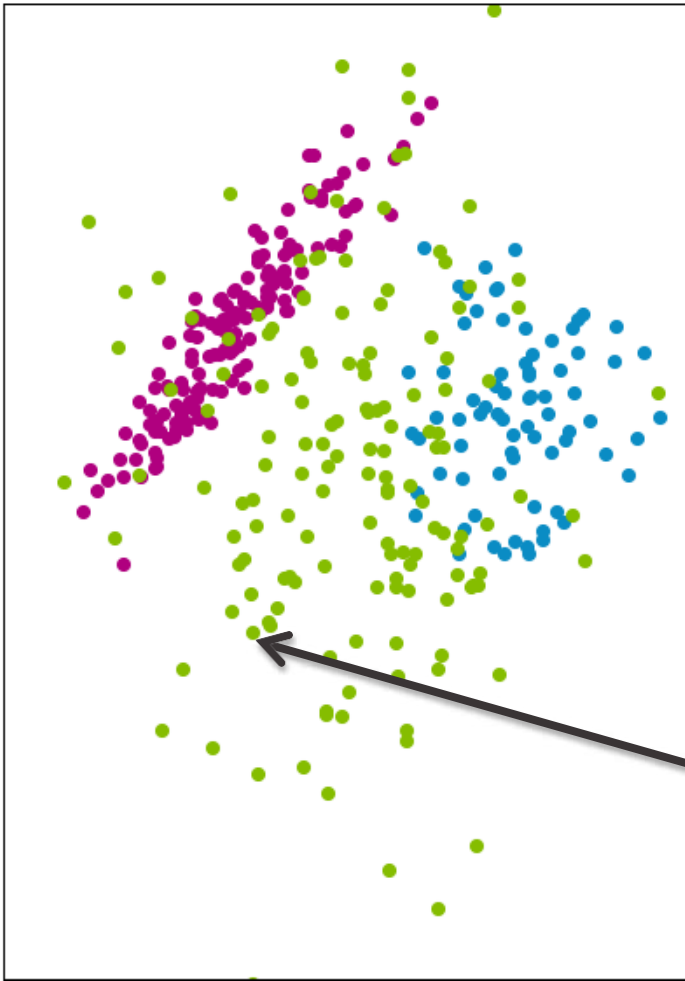
Given observation x_i is from cluster k , what's the likelihood of seeing x_i ? (e.g., just look at distribution for "clouds")

$$p(x_i | z_i = k, \mu_k, \hat{\sigma}_k) = N(x_i | \mu_k, \hat{\sigma}_k)$$



Part 2a:
Imagine we knew the cluster
(hard) assignments z_i

Estimating cluster parameters



Imagine we know the cluster assignments

Estimation problem decouples across clusters

Is green point informative of fuchsia cluster parameters?

NO!

Data table decoupling over clusters

R	G	B	Cluster
$x_1[1]$	$x_1[2]$	$x_1[3]$	3
$x_2[1]$	$x_2[2]$	$x_2[3]$	3
$x_3[1]$	$x_3[2]$	$x_3[3]$	3
$x_4[1]$	$x_4[2]$	$x_4[3]$	1
$x_5[1]$	$x_5[2]$	$x_5[3]$	2
$x_6[1]$	$x_6[2]$	$x_6[3]$	2

Maximum likelihood estimation

R	G	B	Cluster
$x_1[1]$	$x_1[2]$	$x_1[3]$	3
$x_2[1]$	$x_2[2]$	$x_2[3]$	3
$x_3[1]$	$x_3[2]$	$x_3[3]$	3

Estimate $\{\pi_k, \mu_k, \Sigma_k\}$
given data assigned
to cluster k

maximum likelihood estimation
(MLE)

Find parameters that maximize the
score, or *likelihood*, of data

Mean/covariance MLE

R	G	B	Cluster
$x_1[1]$	$x_1[2]$	$x_1[3]$	3
$x_2[1]$	$x_2[2]$	$x_2[3]$	3
$x_3[1]$	$x_3[2]$	$x_3[3]$	3

$$\hat{\mu}_k = \frac{1}{N_k} \sum_{i \text{ in } k} x_i$$

$$\hat{\Sigma}_k = \frac{1}{N_k} \sum_{i \text{ in } k} (x_i - \hat{\mu}_k)(x_i - \hat{\mu}_k)^T$$

Scalar case:

Cluster proportion MLE

R	G	B	Cluster
$x_4[1]$	$x_4[2]$	$x_4[3]$	1

R	G	B	Cluster
$x_5[1]$	$x_5[2]$	$x_5[3]$	2
$x_6[1]$	$x_6[2]$	$x_6[3]$	2

R	G	B	Cluster
$x_1[1]$	$x_1[2]$	$x_1[3]$	3
$x_2[1]$	$x_2[2]$	$x_2[3]$	3
$x_3[1]$	$x_3[2]$	$x_3[3]$	3

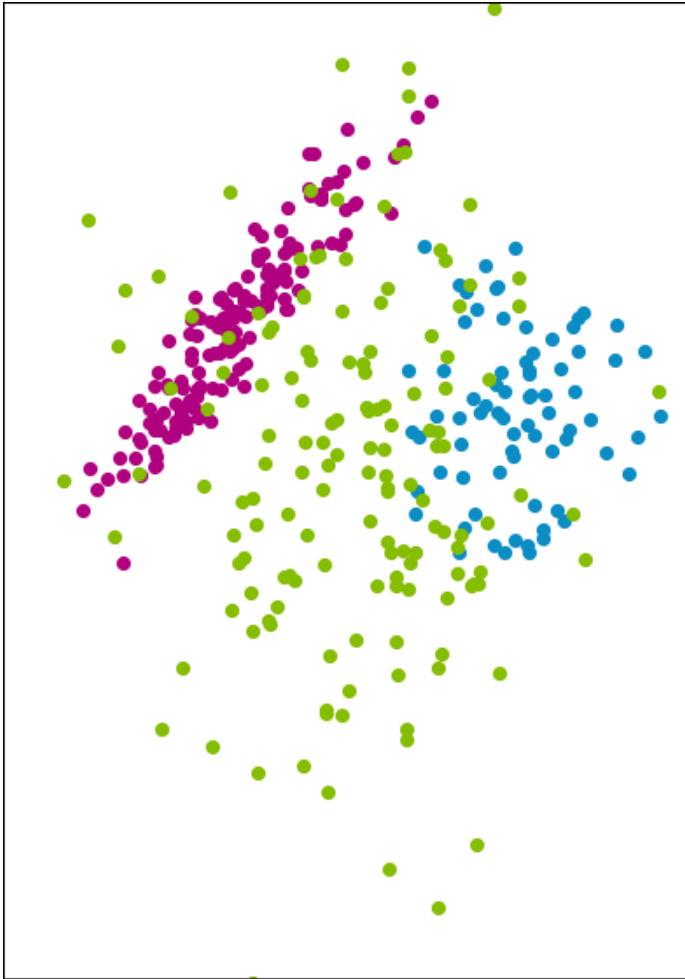
obs in cluster k

$$\uparrow_k = \frac{N_k}{N}$$

total # of obs

True for general mixtures of i.i.d. data,
not just Gaussian clusters

Part 2a summary



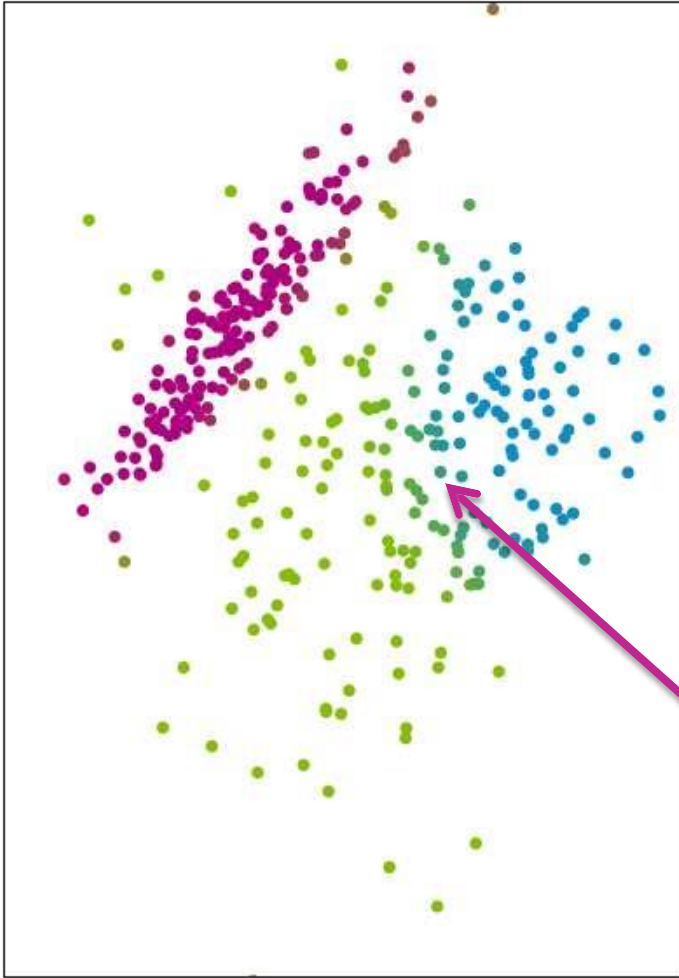
needed to compute soft assignments

Cluster parameters are simple
to compute if we know the
cluster assignments

But, we don't know these!

Part 2b:
**What can we do with just
soft assignments r_{ij} ?**

Estimating cluster parameters from soft assignments



Instead of having a full observation x_i in cluster k , just allocate a portion r_{ik}

x_i divided across all clusters, as determined by r_{ik}

Maximum likelihood estimation from soft assignments

Just like in boosting with weighted observations...

R	G	B	r_{i1}	r_{i2}	r_{i3}
$x_1[1]$	$x_1[2]$	$x_1[3]$	0.30	0.18	0.52
$x_2[1]$	$x_2[2]$	$x_2[3]$	0.01	0.26	0.73
$x_3[1]$	$x_3[2]$	$x_3[3]$	0.002	0.008	0.99
$x_4[1]$	$x_4[2]$	$x_4[3]$	0.75	0.10	0.15
$x_5[1]$	$x_5[2]$	$x_5[3]$	0.05	0.93	0.02
$x_6[1]$	$x_6[2]$	$x_6[3]$	0.13	0.86	0.01

52% chance
this obs is in
cluster 3

Total weight in cluster:
(effective # of obs)

1.242	2.8	2.42
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Maximum likelihood estimation from soft assignments

R	G	B	r_{i1}	r_{i2}	r_{i3}
$x_1[1]$	$x_1[2]$	$x_1[3]$	0.30	0.18	0.52
$x_2[1]$	$x_2[2]$	$x_2[3]$	0.01	0.26	0.73
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$x_6[1]$	$x_6[2]$	$x_6[3]$	0.13	0.86	0.01

Maximum likelihood estimation from soft assignments

R	G	B	Cluster 1 weights		
x ₁ [1]	x ₁ [2]	x ₁ [3]	0.30		
x ₂ [1]	R	G	B	Cluster 2 weights	
x ₃ [1]					
x ₄ [1]	x ₁ [1]	x ₁ [2]	x ₁ [3]	0.18	
x ₅ [1]	x ₂ [1]	R	G	B	Cluster 3 weights
x ₆ [1]	x ₃ [1]				
	x ₄ [1]	x ₁ [1]	x ₁ [2]	x ₁ [3]	0.52
	x ₅ [1]	x ₂ [1]	x ₂ [2]	x ₂ [3]	0.73
	x ₆ [1]	x ₃ [1]	x ₃ [2]	x ₃ [3]	0.99
		x ₄ [1]	x ₄ [2]	x ₄ [3]	0.15
		x ₅ [1]	x ₅ [2]	x ₅ [3]	0.02
		x ₆ [1]	x ₆ [2]	x ₆ [3]	0.01

Cluster-specific location/shape MLE

R	G	B	Cluster 1 weights
$x_1[1]$	$x_1[2]$	$x_1[3]$	0.30
$x_2[1]$	$x_2[2]$	$x_2[3]$	0.01
$x_3[1]$	$x_3[2]$	$x_3[3]$	0.002
$x_4[1]$	$x_4[2]$	$x_4[3]$	0.75
$x_5[1]$	$x_5[2]$	$x_5[3]$	0.05
$x_6[1]$	$x_6[2]$	$x_6[3]$	0.13

$$\hat{\mu}_k = \frac{1}{N_k^{\text{soft}}} \sum_{i=1}^N r_{ik} x_i$$

$$\hat{\Sigma}_k = \frac{1}{N_k^{\text{soft}}} \sum_{i=1}^N r_{ik} (x_i - \hat{\mu}_k)(x_i - \hat{\mu}_k)^T$$

$$N_k^{\text{soft}} = \sum_{i=1}^N r_{ik}$$

Total weight in cluster k
= effective # obs

Compute cluster parameter estimates
with weights on each row operation

MLE of cluster proportions $\hat{\pi}_k$

r_{i1}	r_{i2}	r_{i3}
0.30	0.18	0.52
0.01	0.26	0.73
0.002	0.008	0.99
0.75	0.10	0.15
0.05	0.93	0.02
0.13	0.86	0.01

Total weight
in cluster:

1.242	2.8	2.42
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Total weight
in dataset:

6

datapoints N

$$\hat{\pi}_k = \frac{N_k^{\text{soft}}}{N}$$

Estimate cluster
proportions from
relative weights

$$N_k^{\text{soft}} = \sum_{i=1}^N r_{ik}$$

Total weight in cluster k
= effective # obs

Defaults to hard assignment case when r_{ij} in $\{0,1\}$

Hard assignments have:

$$r_{ik} = \begin{cases} 1 & i \text{ in } k \\ 0 & \text{otherwise} \end{cases}$$

R	G	B	r_{i1}	r_{i2}	r_{i3}
$x_1[1]$	$x_1[2]$	$x_1[3]$	0	0	1
$x_2[1]$	$x_2[2]$	$x_2[3]$	0	0	1
$x_3[1]$	$x_3[2]$	$x_3[3]$	0	0	1
$x_4[1]$	$x_4[2]$	$x_4[3]$	1	0	0
$x_5[1]$	$x_5[2]$	$x_5[3]$	0	1	0
$x_6[1]$	$x_6[2]$	$x_6[3]$	0	1	0

One-hot encoding of cluster assignment

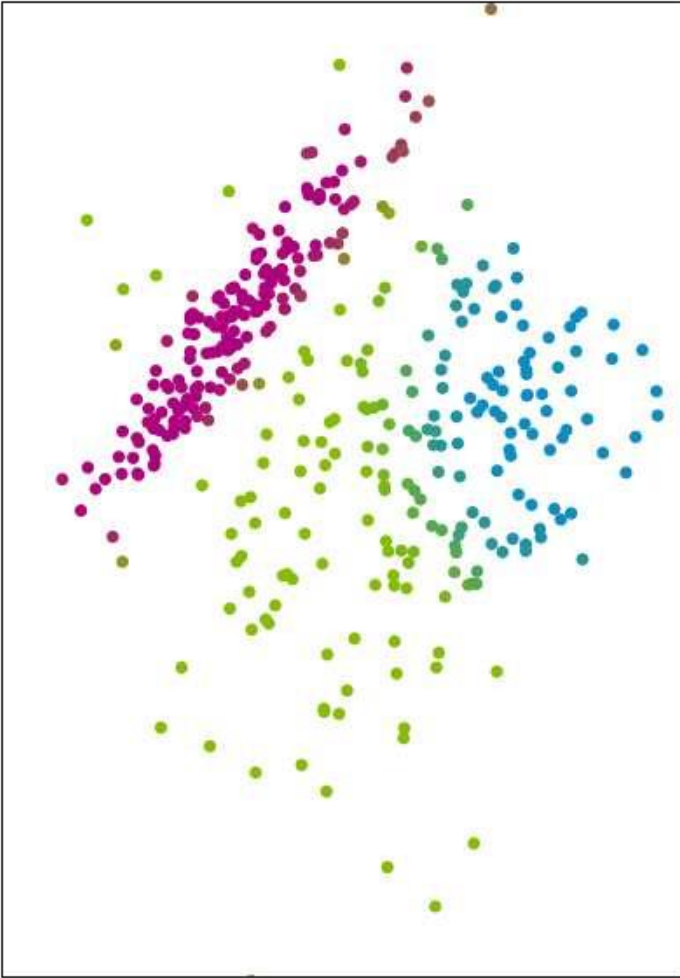
Total weight in cluster:

1	2	3
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Equating the estimates...

$$\begin{aligned}\hat{\pi}_k &= \frac{N_k^{\text{soft}}}{N} \quad \leftarrow \boxed{N_k^{\text{soft}} = \sum_{i=1}^N r_{ik}} \\ \hat{\mu}_k &= \frac{1}{N_k^{\text{soft}}} \sum_{i=1}^N r_{ik} x_i \\ \hat{\Sigma}_k &= \frac{1}{N_k^{\text{soft}}} \sum_{i=1}^N r_{ik} (x_i - \hat{\mu}_k)(x_i - \hat{\mu}_k)^T\end{aligned}$$

Part 2b summary



Still straightforward
to compute cluster
parameter estimates
from soft assignments

Expectation maximization (EM)

Expectation maximization (EM): An iterative algorithm

Motivates an iterative algorithm:

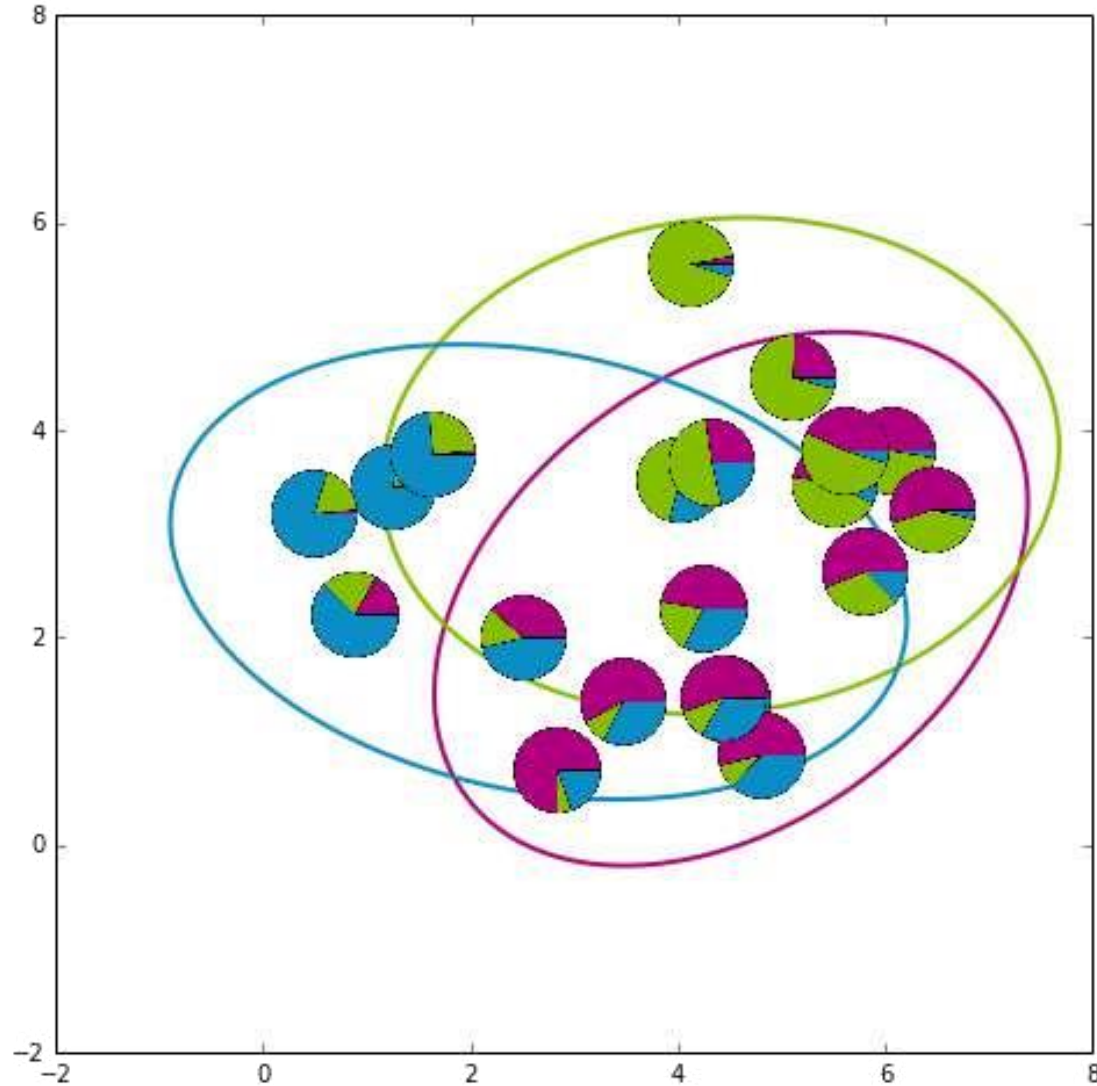
1. E-step: estimate cluster responsibilities given current parameter estimates

$$\hat{r}_{ik} = \frac{\hat{\pi}_k N(x_i | \hat{\mu}_k, \hat{\Sigma}_k)}{\sum_{j=1}^K \hat{\pi}_j N(x_i | \hat{\mu}_j, \hat{\Sigma}_j)}$$

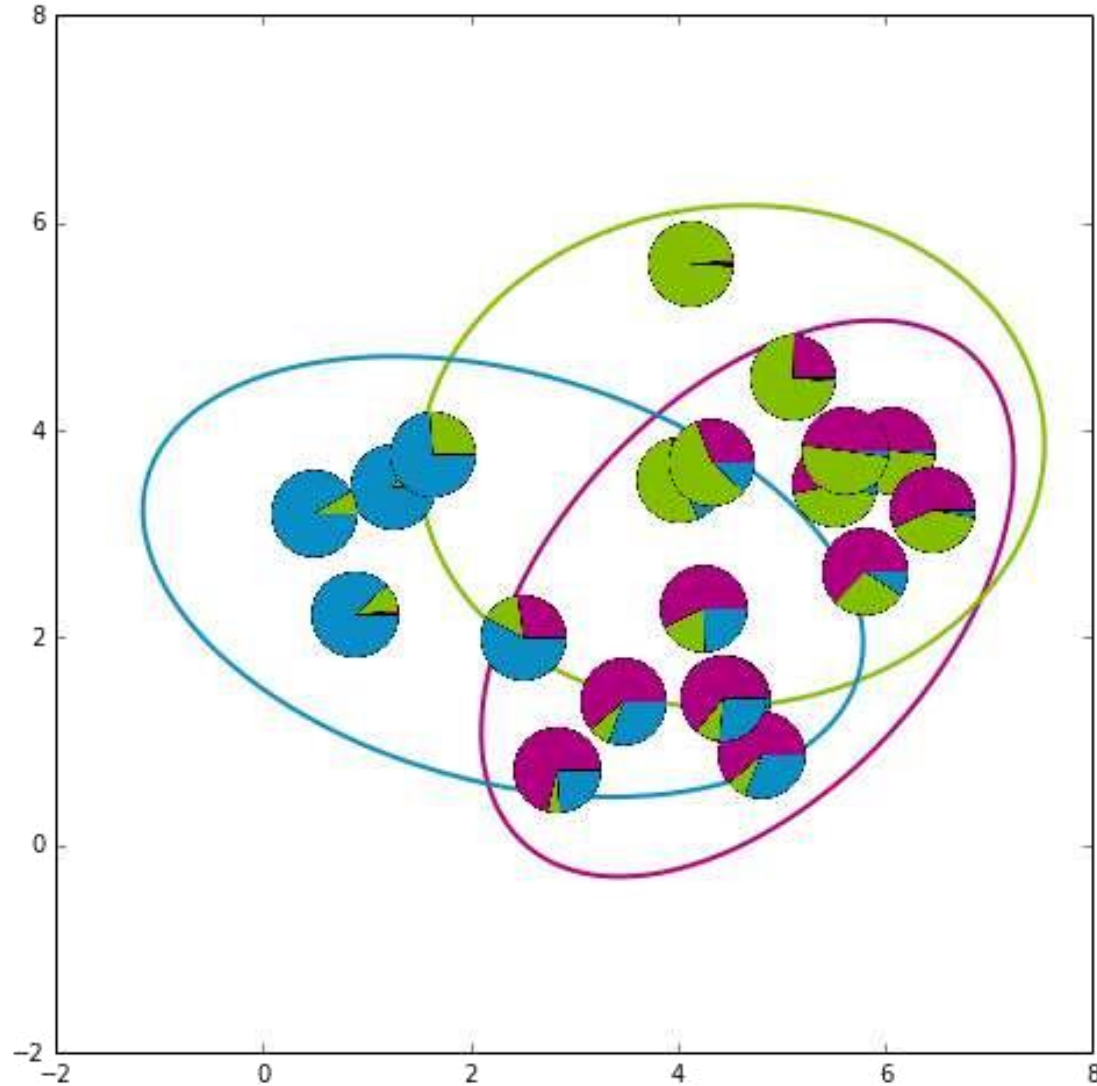
2. M-step: maximize likelihood over parameters given current responsibilities

$$\hat{\pi}_k, \hat{\mu}_k, \hat{\Sigma}_k \mid \{\hat{r}_{ik}, x_i\}$$

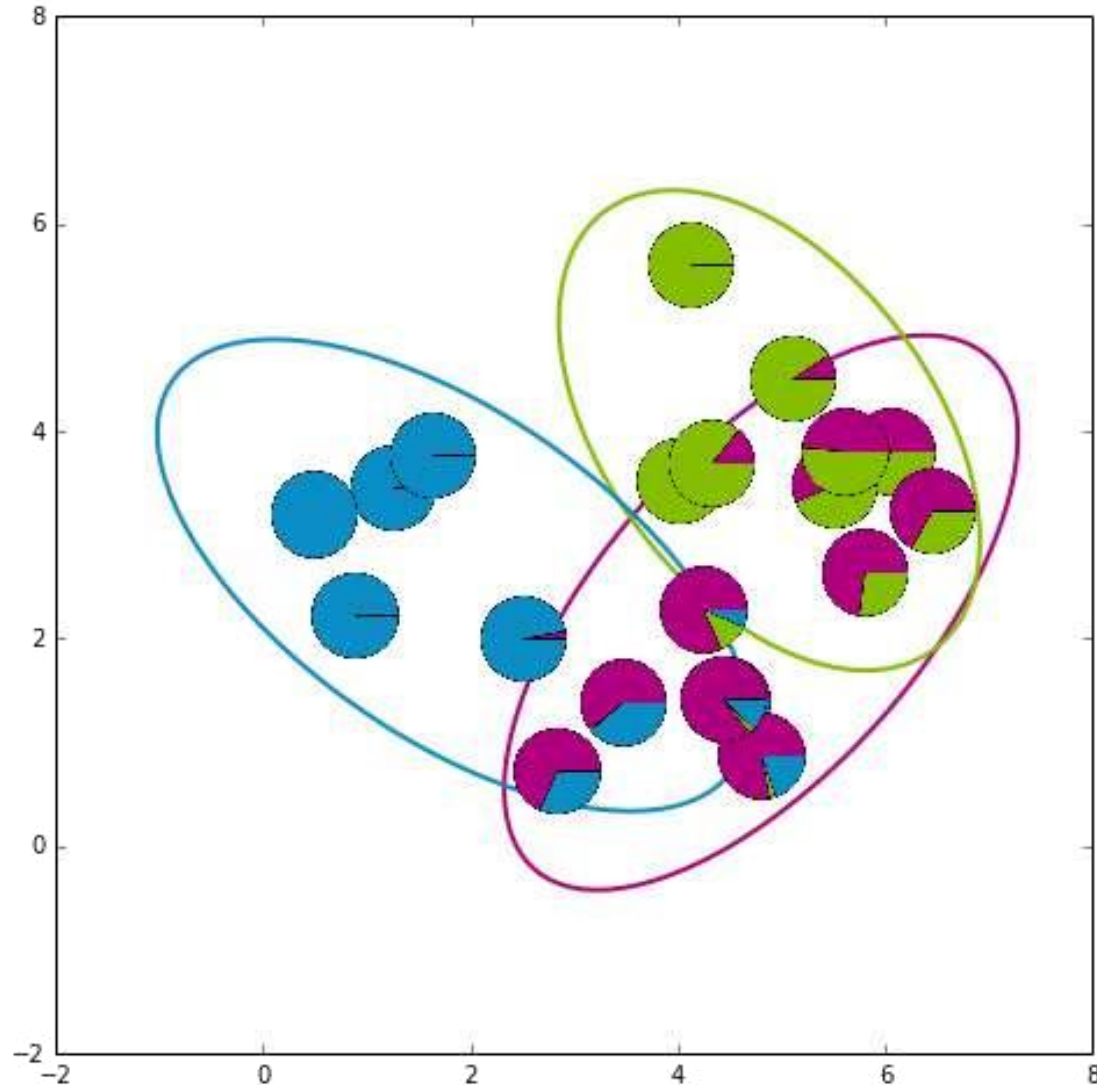
EM for mixtures of Gaussians in pictures – initialization



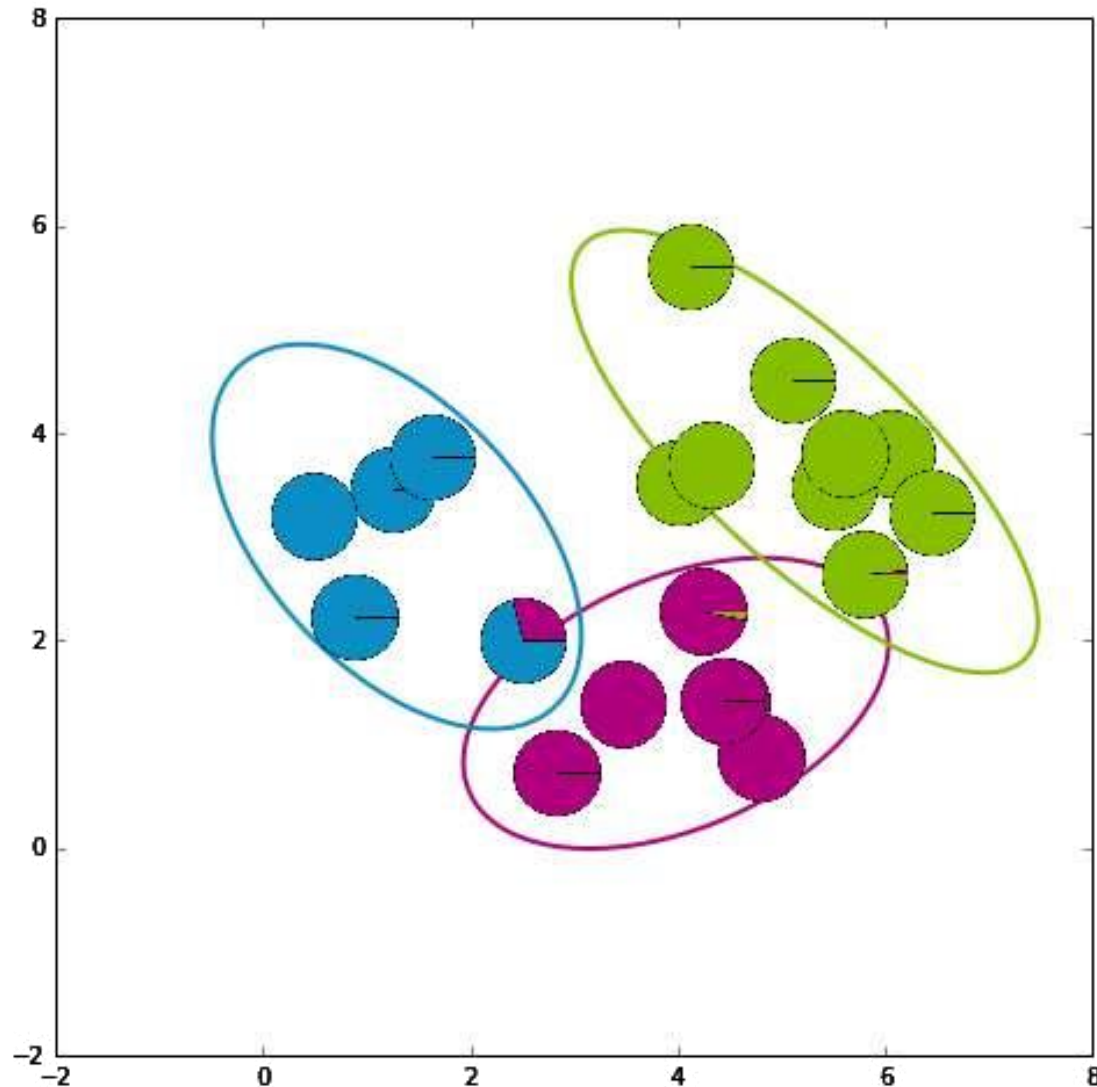
EM for mixtures of Gaussians in pictures – after 1st iteration



EM for mixtures of Gaussians in pictures – after 2nd iteration



EM for mixtures of Gaussians in pictures – converged solution



The nitty gritty of EM

Convergence of EM

- EM is a **coordinate-ascent algorithm**
 - Can equate E-and M-steps with alternating maximizations of an objective function
- Converges to a **local mode**
- We will assess via (log) likelihood of data under current parameter and responsibility estimates

Initialization

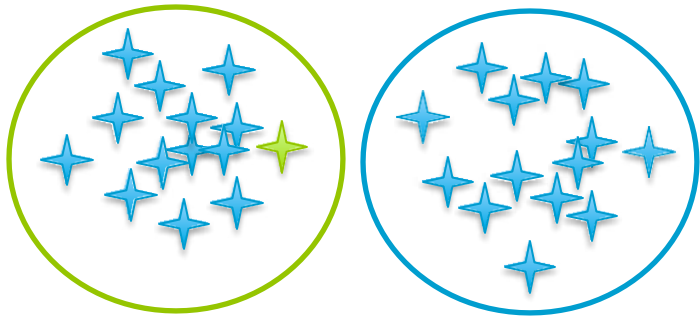
- Many ways to initialize the EM algorithm
- Important for convergence rates and quality of local mode found
- Examples:
 - Choose K observations at random to define K “centroids”. Assign other observations to nearest centroid to form initial parameter estimates.
 - Pick centers sequentially to provide good coverage of data like in k-means++
 - Initialize from k-means solution
 - Grow mixture model by splitting (and sometimes removing) clusters until K clusters are formed

Overfitting of MLE

Maximizing likelihood can **overfit to data**

Imagine at $K=2$ example with one obs assigned to **cluster 1** and others assigned to **cluster 2**

- What parameter values maximize likelihood?



Set center equal to
point and shrink
variance to 0

Likelihood goes to ∞ !

Overfitting in high dims

Doc-clustering example:

Imagine only 1 doc assigned to cluster k has word w
(or all docs in cluster agree on count of word w)

Likelihood maximized by setting $\mu_k[w] = x_i[w]$ and $\sigma_{w,k}^2 = 0$

Likelihood of any doc with different count on
word w being in cluster k is 0!

Simple regularization of M-step for mixtures of Gaussians

Simple fix: Don't let variances $\rightarrow 0$!

Add small amount to diagonal of covariance estimate

Alternatively, take Bayesian approach and place prior on parameters.

Similar idea, but all parameter estimates are “smoothed” via cluster pseudo-observations.

Relationship to k-means

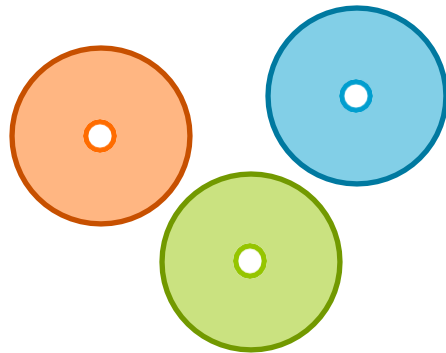
Consider Gaussian mixture model with

$$\Sigma = \begin{pmatrix} \sigma^2 & & & \\ & \sigma^2 & & \\ & & \sigma^2 & \\ & & & \ddots \\ & & & & \sigma^2 \end{pmatrix}$$

and let the variance parameter $\sigma \rightarrow 0$

Datapoint gets fully assigned to nearest center, just as in k-means

Spherically
symmetric clusters



- Spherical clusters with equal variances, so **relative likelihoods** just function of distance to cluster center
- As variances $\rightarrow 0$, **likelihood ratio becomes 0 or 1**
- Responsibilities weigh in cluster proportions, but dominated by likelihood disparity

$$\hat{r}_{ik} = \frac{\hat{\pi}_k N(x_i | \hat{\mu}_k, \sigma^2 I)}{\sum_{j=1}^K \hat{\pi}_j N(x_i | \hat{\mu}_j, \sigma^2 I)}$$

Infinitesimally small variance EM = k-means

1. E-step: estimate cluster responsibilities given current parameter estimates

$$\hat{r}_{ik} = \frac{\hat{\pi}_k N(x_i \mid \hat{\mu}_k, \sigma^2 I)}{\sum_{j=1}^K \hat{\pi}_j N(x_i \mid \hat{\mu}_j, \sigma^2 I)} \in \{0, 1\}$$

Infinitesimally small

Decision based on
distance to nearest
cluster center

2. M-step: maximize likelihood over parameters given current responsibilities (**hard assignments!**)

$$\hat{\pi}_k, \hat{\mu}_k \mid \{\hat{r}_{ik}, x_i\}$$

Let's Work...

[Colab Notebook](https://colab.research.google.com/drive/1gDgPzuqwgGOcNBIF_D4RA7izSiW7f71y?usp=sharing)

https://colab.research.google.com/drive/1gDgPzuqwgGOcNBIF_D4RA7izSiW7f71y?usp=sharing

Download it first then upload in colab platform

Thank you

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